VOTARYTECH
Mobility. Solutions. Services

2 Wheeler Cluster Display

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Project Objective & System Overview

- Modern vehicles are rapidly shifting from traditional analog meters to fully digital instrument clusters. Digital displays provide **better visibility, real-time updates, and enhanced user interaction**.
- This project aims to develop a **digital instrument cluster** using **Qt/QML** on an **embedded platform**, capable of displaying **real-time vehicle parameters** received wirelessly from an **ESP32 sensor module** through **Wi-Fi and Firebase**.

Implemented Features

- Displays **real-time speed and temperature** fetched from ESP32 via **Wi-Fi and Firebase**
- Shows **current time and date** using the OKT507-C RTC
- Provides a **graphical battery level indicator**
- Includes **Anti-Theft Lock** for vehicle security
- Detects **vehicle tilt/fall** and triggers an alarm
- Implements a **Wi-Fi configuration GUI** for hotspot connection

System Architecture (Working Principle)

- **ESP32** acts as the sensor data acquisition unit
- Sensor data is transmitted wirelessly using **Wi-Fi**
- **Firebase** is used as a cloud intermediary for data transfer
- **C++ backend** on OKT507-C handles data fetching and processing
- **QML UI layer** displays the received data in real time on the screen

Hardware Components Used

ESP32 Microcontroller

- Used as the microcontroller to sense the data from the sensors
- Collects real-time speed and temperature values
- Provides built-in Wi-Fi connectivity for data transmission to firebase

OKT507-C Embedded Board

- Acts as the processing and display unit
- Runs **Embedded Linux**
- Executes the Qt/QML digital instrument cluster application

Display Panel

- Visualizes speed, temperature, time/date, battery level, settings features and Anti-Theft.

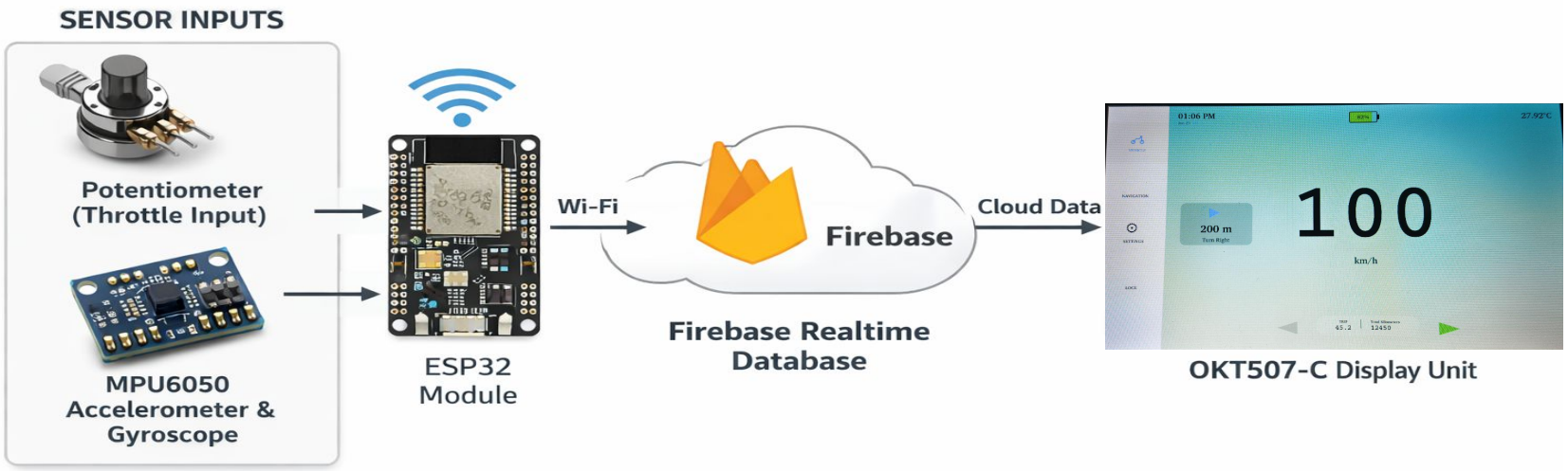
Power Supply

- Provides regulated power to ESP32 and OKT507-C board

Tools, Kits & Languages Used

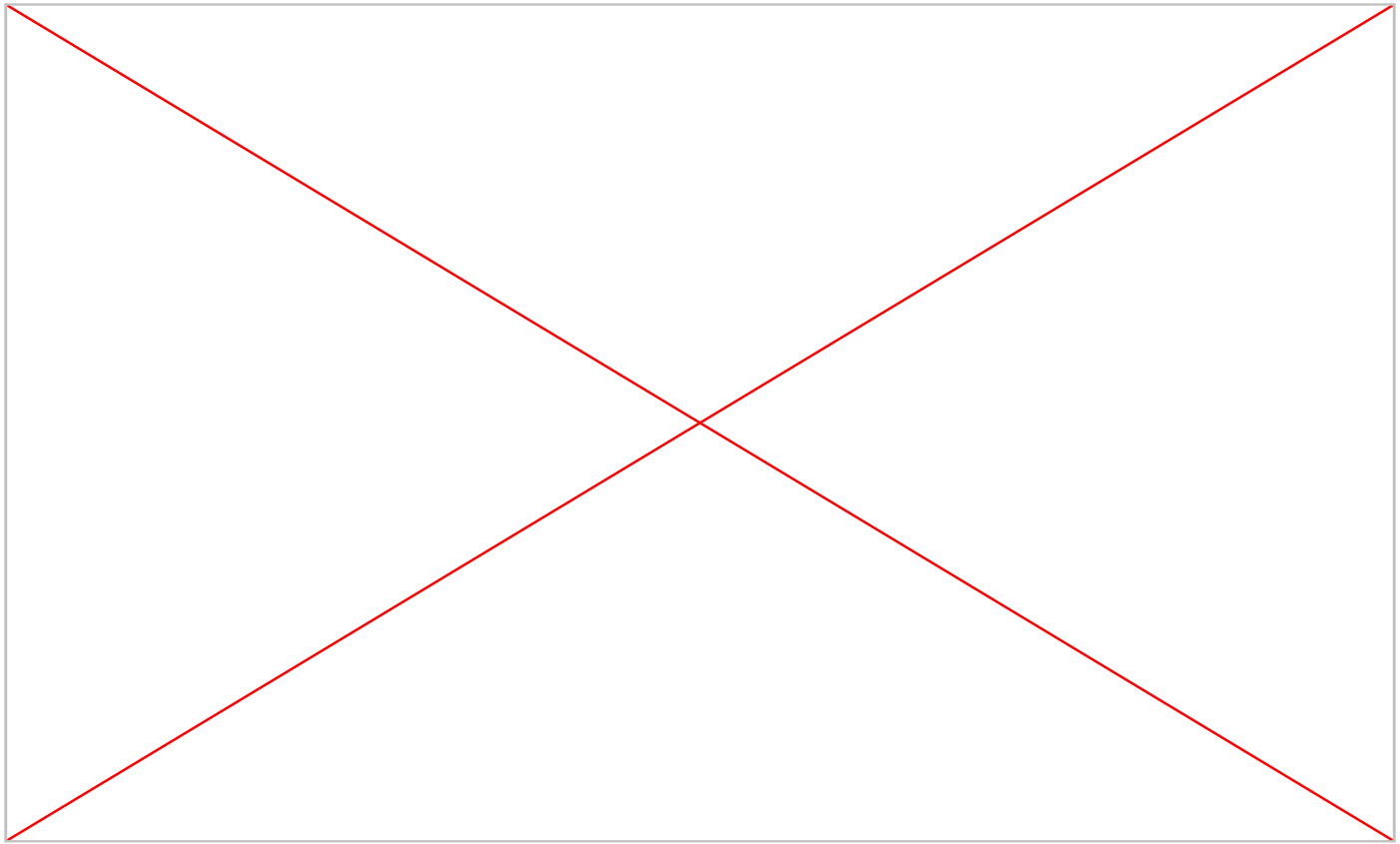
- **Development Environment**
  - Ubuntu Linux
  - Qt Creator
  - Arduino IDE
- **Embedded Linux**
  - Custom Linux image on OKT507-C
- **Hardware Platform**
  - OKT507-C Single Board Computer
  - External ESP32 sensor module
- **Programming Languages**
  - **C++** – backend logic and data handling
  - **QML** – real-time UI and animations
  - **Python Script** – To bypass firebase data to PC and to OKT

WorkFlow



Sensor Inputs → ESP32 Module → Firebase Cloud → Embedded Display

2 Wheeler Cluster Display



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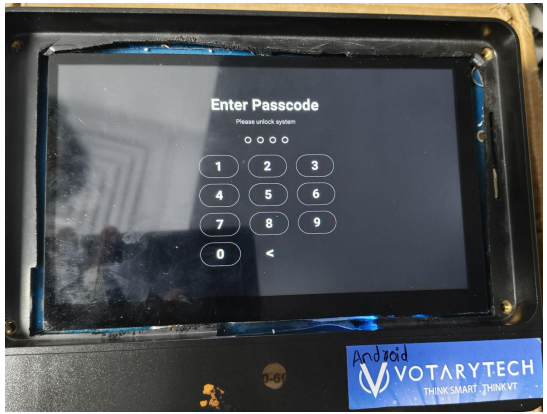

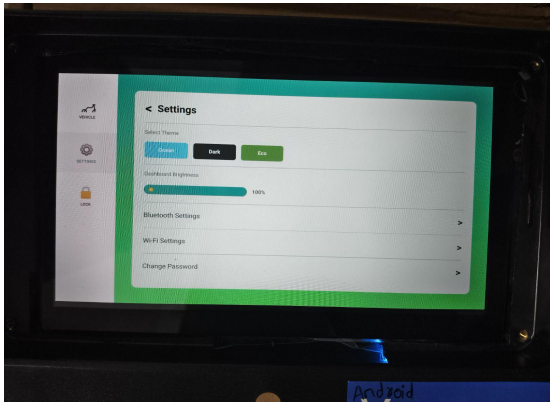

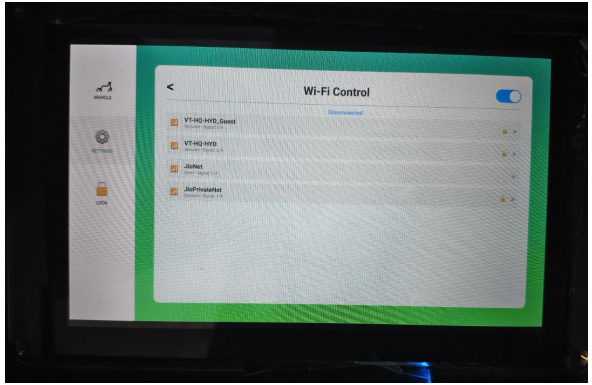

Automotive Industry Standards vs. OKT507-C Implementation

Aspect	Industry-Standard (2-Wheeler Cluster)	Implemented Solution (OKT507-C)
Primary Controller	Automotive-Grade SoC (e.g., NXP i.MX, Renesas) or MCU (STM32/Infineon)	Industrial-Grade SBC (Forlinx OKT507-C)
Operating System	Real-Time OS (QNX, FreeRTOS) or Android Automotive	Embedded Linux (Standard Distribution)
Data Communication	CAN Bus / LIN Bus (Wired, Deterministic, Low Latency)	Wi-Fi + HTTP Bridge (Wireless, Cloud-Dependent)
Sensor Interface	Hardwired to ECU/BCM (Body Control Module)	Wireless IoT Node (ESP32) via Firebase Cloud
Graphics Engine	Qt Automotive Suite / Kanzi UI / Altia	Qt 5.12 / QML (Community Edition)
Data Gateway	Central Gateway ECU (Secure Onboard Comms)	Python Script Bridge (PC-based Data Forwarding)
Theft Protection	Hardware Immobilizer (Key Transponder / ECU Lock)	Software Anti-Theft (PIN/Pattern Lock Screen)
Safety Features	ABS / TCS / Side-Stand Cutoff (Hardwired Signals)	Tilt/Fall Detection (Accelerometer via ESP32)
Boot Time	Instant (< 2 seconds required by regulation)	Linux Boot Sequence (~15â€“30 seconds)
Power Management	Automotive PMIC (Handles 12V cranks/surges)	Regulated 5V DC Supply (Stable Bench Power)
Connectivity	Bluetooth (Phone Pairing) / LTE (Telematics)	Wi-Fi (Hotspot connection for Data Sync)

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2 Wheeler EV Cluster (Ather)	OKT 2Wheeler Cluster

THANK YOU